Step 2 : Data

Instructor 1	2	3	5	0	8		
Instructor 2	4	6	8	4	9	0	2
Instructor 3	5	2	3	2	3	3	

Step 3 : Level of significance $\alpha = 5\%$

Step 4 : Test Statistic

$$F_0 = \frac{MST}{MSE}$$

Step 5 : Calculation of Test statistic

								Total	Square
Instructor 1	2	3	5	0	8			18	324
Instructor 2	4	6	8	4	9	0	2	33	1089
Instructor 3	5	2	3	2	3	3		18	324
								69	

Individual squares

Instructor 1	4	9	25	0	64		
Instructor 2	16	36	64	16	81	0	4
Instructor 3	25	4	9	4	9	9	

$$\sum\sum x_{ij}^2 = 379$$

Correction Factor: $CF = \frac{G^2}{n} = \frac{(69)^2}{18} = \frac{4761}{18} = 264.5$

Total Sum of Squares: $TSS = \sum\sum x_{ij}^2 - C.F$

$= 379 - 264.5 = 114.5$

Sum of Squares between Treatments: $SST = \frac{\sum x_i^2}{n_i} - C.F$

$= \left(\frac{324}{5} + \frac{1089}{7} + \frac{324}{6}\right) - 264.5$

$= (64.8 + 155.6 + 54) - 264.5$

$= (274.4) - 264.5$

$= 9.9$

Sum of Squares due to Error: $SSE = TSS - SST$

$= 114.5 - 9.9$

$= 104.6$

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